**Problem 6.6.** Of the following materials, which would you expect to be paramagnetic and which diamagnetic: aluminum, copper, copper chloride ($CuCl_2$), carbon, lead, nitrogen ($N_2$), salt ($NaCl$), sodium, sulfur, water? (Actually, copper is slightly *dia*magnetic; otherwise they're all what you'd expect.)[8]

## 6.2 The Field of a Magnetized Object

### 6.2.1 Bound Currents

Suppose we have a piece of magnetized material; the magnetic dipole moment per unit volume, **M**, is given. What field does this object produce? Well, the vector potential of a single dipole **m** is given by Eq. 5.85:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{\boldsymbol{\imath}}}{\imath^2}. \tag{6.10}$$

In the magnetized object, each volume element $d\tau'$ carries a dipole moment $\mathbf{M}\,d\tau'$, so the total vector potential is (Fig. 6.11)

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{M}(\mathbf{r}')\times\hat{\boldsymbol{\imath}}}{\imath^2}\,d\tau. \tag{6.11}$$

That *does* it, in principle. But, as in the electrical case (Section 4.2.1), the integral can be cast in a more illuminating form by exploiting the identity

$$\boldsymbol{\nabla}'\frac{1}{\imath} = \frac{\hat{\boldsymbol{\imath}}}{\imath^2}.$$

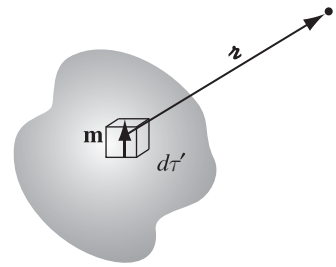



Fig. 6.11

[8] In truth, the determination is more complicated than I have suggested. Although an individual atom (or molecule) with an odd number of electrons obviously has an unpaired electron, and is therefore paramagnetic, those with an even number can also have unpaired electrons (this is governed by **Hund's rule**). Atomic oxygen, for example, is paramagnetic. Moreover, in molecules and solids, pairing may occur between *shared* electrons – thus bismuth (atomic number 83) is strongly diamagnetic in its solid phase.

With this,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \left[\mathbf{M}(\mathbf{r}') \times \left(\boldsymbol{\nabla}' \frac{1}{\mathcal{r}}\right)\right] d\tau'.$$

Integrating by parts, using product rule 7, gives

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \left\{ \int \frac{1}{\mathcal{r}} [\boldsymbol{\nabla}' \times \mathbf{M}(\mathbf{r}')]\, d\tau' - \int \boldsymbol{\nabla}' \times \left[\frac{\mathbf{M}(\mathbf{r}')}{\mathcal{r}}\right] d\tau' \right\}.$$

Problem 1.61b invites us to express the latter as a surface integral,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{1}{\mathcal{r}} [\boldsymbol{\nabla}' \times \mathbf{M}(\mathbf{r}')]\, d\tau' + \frac{\mu_0}{4\pi} \oint \frac{1}{\mathcal{r}} [\mathbf{M}(\mathbf{r}') \times d\mathbf{a}']. \tag{6.12}$$

The first integral looks just like the potential of a volume current,

$$\boxed{\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M},} \tag{6.13}$$

while the second looks like the potential of a surface current,

$$\boxed{\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}},} \tag{6.14}$$

where $\hat{\mathbf{n}}$ is the normal unit vector. With these definitions,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{\mathcal{V}} \frac{\mathbf{J}_b(\mathbf{r}')}{\mathcal{r}}\, d\tau' + \frac{\mu_0}{4\pi} \oint_{\mathcal{S}} \frac{\mathbf{K}_b(\mathbf{r}')}{\mathcal{r}}\, da'. \tag{6.15}$$

What this means is that the potential (and hence also the field) is the same as would be produced by a volume current $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$ throughout the material, plus a surface current $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$, on the boundary. Instead of integrating the contributions of all the infinitesimal dipoles, using Eq. 6.11, we first determine the **bound currents**, and then find the field *they* produce, in the same way we would calculate the field of any other volume and surface currents. Notice the striking parallel with the electrical case: there the field of a polarized object was the same as that of a bound volume charge $\rho_b = -\boldsymbol{\nabla} \cdot \mathbf{P}$ plus a bound surface charge $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$.

**Example 6.1.** Find the magnetic field of a uniformly magnetized sphere.

*Solution.* Choosing the $z$-axis along the direction of $\mathbf{M}$ (Fig. 6.12), we have

$$\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M} = \mathbf{0}, \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = M \sin\theta\, \hat{\boldsymbol{\phi}}.$$

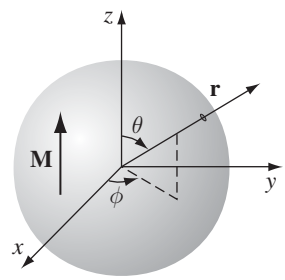



Fig. 6.12

Now, a rotating spherical shell, of uniform surface charge $\sigma$, corresponds to a surface current density

$$\mathbf{K} = \sigma \mathbf{v} = \sigma \omega R \sin\theta \, \hat{\boldsymbol{\phi}}.$$

It follows, therefore, that the field of a uniformly magnetized sphere is identical to the field of a spinning spherical shell, with the identification $\sigma R\boldsymbol{\omega} \to \mathbf{M}$. Referring back to Ex. 5.9,

$$\mathbf{B} = \frac{2}{3}\mu_0 \mathbf{M}, \tag{6.16}$$

inside the sphere, while the field outside is the same as that of a perfect dipole,

$$\mathbf{m} = \frac{4}{3}\pi R^3 \mathbf{M}.$$

Notice that the internal field is *uniform*, like the electric field inside a uniformly polarized sphere (Eq. 4.14), although the actual *formulas* for the two cases are curiously different ($\frac{2}{3}$ in place of $-\frac{1}{3}$).[9] The external fields are also analogous: pure dipole in both instances.

**Problem 6.7.** An infinitely long circular cylinder carries a uniform magnetization $\mathbf{M}$ parallel to its axis. Find the magnetic field (due to $\mathbf{M}$) inside and outside the cylinder.

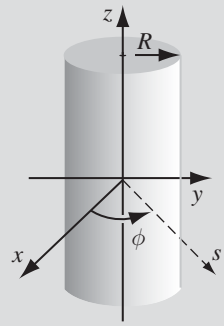



Fig. 6.13

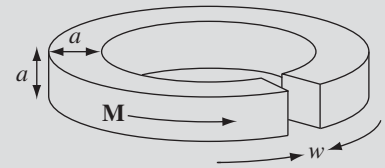



Fig. 6.14

**Problem 6.8.** A very long circular cylinder of radius $R$ carries a magnetization $\mathbf{M} = ks^2\hat{\boldsymbol{\phi}}$, where $k$ is a constant (Fig. 6.13). Find the magnetic field due to $\mathbf{M}$, for points inside and outside the cylinder (and far from the ends).

**Problem 6.9.** A short circular cylinder of radius $a$ and length $L$ carries a "frozen-in" uniform magnetization $\mathbf{M}$ parallel to its axis. Find the bound current, and sketch the magnetic field of the cylinder. (Make three sketches: one for $L \gg a$, one for $L \ll a$, and one for $L \approx a$.) Compare this **bar magnet** with the bar electret of Prob. 4.11.

[9] It is no accident that the same factors appear in the "contact" term for the fields of electric and magnetic dipoles (Eqs. 3.106 and 5.94). In fact, one good way to model a perfect dipole is to take the limit ($R \to 0$) of a uniformly polarized/magnetized sphere.

**Problem 6.10.** An iron rod of length $L$ and square cross section (side length $a$) is given a uniform longitudinal magnetization $\mathbf{M}$, and then bent around into a circle with a narrow gap (width $w$), as shown in Fig. 6.14. Find the magnetic field at the center of the gap, assuming $w \ll a \ll L$. [*Hint:* Treat it as the superposition of a *complete* torus plus a square loop with reversed current.]

### 6.2.2 Physical Interpretation of Bound Currents

In the last section, we found that the field of a magnetized object is identical to the field that would be produced by a certain distribution of "bound" currents, $\mathbf{J}_b$ and $\mathbf{K}_b$. I want to show you how these bound currents arise physically. This will be a heuristic argument – the rigorous derivation has already been given. Figure 6.15 depicts a thin slab of uniformly magnetized material, with the dipoles represented by tiny current loops. Notice that all the "internal" currents cancel: every time there is one going to the right, a contiguous one is going to the left. However, at the edge there is *no adjacent loop to do the canceling*. The whole thing, then, is equivalent to a single ribbon of current $I$ flowing around the boundary (Fig. 6.16).

What *is* this current, in terms of $\mathbf{M}$? Say that each of the tiny loops has area $a$ and thickness $t$ (Fig. 6.17). In terms of the magnetization $M$, its dipole moment is $m = Mat$. In terms of the circulating current $I$, however, $m = Ia$. Therefore $I = Mt$, so the surface current is $K_b = I/t = M$. Using the

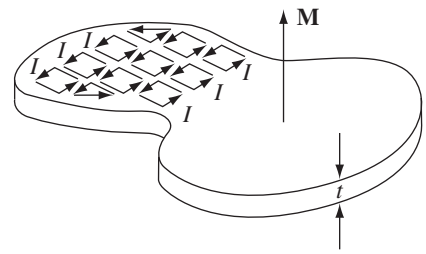



Fig. 6.15

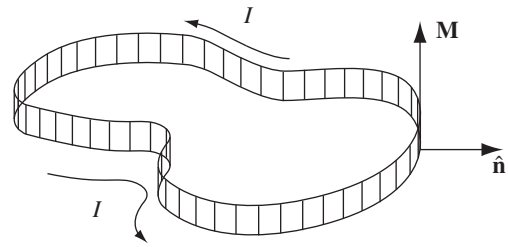



Fig. 6.16

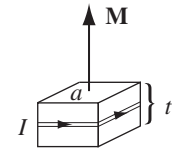



Fig. 6.17

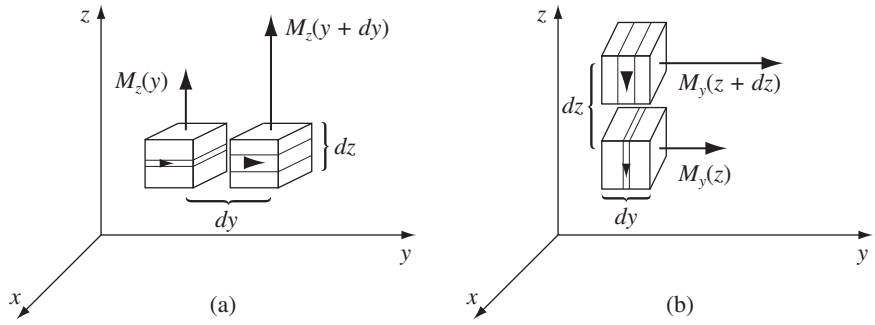



Fig. 6.18

outward-drawn unit vector $\hat{\mathbf{n}}$ (Fig. 6.16), the direction of $\mathbf{K}_b$ is conveniently indicated by the cross product

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}.$$

(This expression also records the fact that there is *no* current on the top or bottom surface of the slab; here $\mathbf{M}$ is parallel to $\hat{\mathbf{n}}$, so the cross product vanishes.)

This bound surface current is exactly what we obtained in Section 6.2.1. It is a peculiar *kind* of current, in the sense that no single charge makes the whole trip – on the contrary, each charge moves only in a tiny loop within a single atom. Nevertheless, the net effect is a macroscopic current flowing over the surface of the magnetized object. We call it a "bound" current to remind ourselves that every charge is attached to a particular atom, but it's a perfectly genuine current, and it produces a magnetic field in the same way any other current does.

When the magnetization is *non*uniform, the internal currents no longer cancel. Figure 6.18a shows two adjacent chunks of magnetized material, with a larger arrow on the one to the right suggesting greater magnetization at that point. On the surface where they join, there is a net current in the $x$ direction, given by

$$I_x = [M_z(y + dy) - M_z(y)]\, dz = \frac{\partial M_z}{\partial y}\, dy\, dz.$$

The corresponding volume current density is therefore

$$(J_b)_x = \frac{\partial M_z}{\partial y}.$$

By the same token, a nonuniform magnetization in the $y$ direction would contribute an amount $-\partial M_y/\partial z$ (Fig. 6.18b), so

$$(J_b)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}.$$

In general, then,

$$\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M},$$

consistent, again, with the result of Section 6.2.1.

Incidentally, like any other steady current, $\mathbf{J}_b$ should obey the conservation law 5.33:

$$\boldsymbol{\nabla} \cdot \mathbf{J}_b = 0.$$

Does it? *Yes*; the divergence of a curl is *always* zero.

### 6.2.3 The Magnetic Field inside Matter

Like the electric field, the true *microscopic* magnetic field inside matter fluctuates wildly from point to point and instant to instant. When we speak of "the" magnetic field in matter, we mean the *macroscopic* field: the average over regions large enough to contain many atoms. (The magnetization $\mathbf{M}$ is "smoothed out" in the same sense.) It is this macroscopic field that one obtains when the methods of Section 6.2.1 are applied to points inside magnetized material, as you can prove for yourself in the following problem.

**Problem 6.11.** In Sect, 6.2.1, we began with the potential of a *perfect* dipole (Eq. 6.10), whereas in *fact* we are dealing with *physical* dipoles. Show, by the method of Section 4.2.3, that we nonetheless get the correct macroscopic field.

## 6.3 The Auxiliary Field $\mathbf{H}$

### 6.3.1 Ampère's Law in Magnetized Materials

In Section 6.2, we found that the effect of magnetization is to establish bound currents $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$ within the material and $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ on the surface. The field due to magnetization of the medium is just the field produced by these bound currents. We are now ready to put everything together: the field attributable to bound currents, plus the field due to everything else – which I shall call the **free current**. The free current might flow through wires embedded in the magnetized substance or, if the latter is a conductor, through the material itself. In any event, the total current can be written as

$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f. \tag{6.17}$$

There is no new physics in Eq. 6.17; it is simply a *convenience* to separate the current into these two parts, because they got there by quite different means: the free current is there because somebody hooked up a wire to a battery – it involves actual transport of charge; the bound current is there because of magnetization – it results from the conspiracy of many aligned atomic dipoles.

In view of Eqs. 6.13 and 6.17, Ampère's law can be written

$$\frac{1}{\mu_0}(\boldsymbol{\nabla} \times \mathbf{B}) = \mathbf{J} = \mathbf{J}_f + \mathbf{J}_b = \mathbf{J}_f + (\boldsymbol{\nabla} \times \mathbf{M}),$$

or, collecting together the two curls,

$$\boldsymbol{\nabla} \times \left(\frac{1}{\mu_0}\mathbf{B} - \mathbf{M}\right) = \mathbf{J}_f.$$

The quantity in parentheses is designated by the letter **H**:

$$\boxed{\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}.} \tag{6.18}$$

In terms of **H**, then, Ampère's law reads

$$\boxed{\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}_f,} \tag{6.19}$$

or, in integral form,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{f\text{enc}}, \tag{6.20}$$

where $I_{f\text{enc}}$ is the total *free* current passing through the Ampèrian loop.

The vector **H** plays a role in magnetostatics analogous to **D** in electrostatics: just as **D** allowed us to write *Gauss's* law in terms of the free *charge* alone, **H** permits us to express *Ampère's* law in terms of the free *current* alone – and free current is what we control directly. Bound current, like bound charge, comes along for the ride – the material gets magnetized, and this results in bound currents; we cannot turn them on or off independently, as we can free currents. In applying Eq. 6.20, all we need to worry about is the *free* current, which we know about because we *put* it there. In particular, when symmetry permits, we can calculate **H** immediately from Eq. 6.20 by the usual Ampère's law methods. (For example, Probs. 6.7 and 6.8 can be done in one line by noting that $\mathbf{H} = \mathbf{0}$.)

**Example 6.2.** A long copper rod of radius $R$ carries a uniformly distributed (free) current $I$ (Fig. 6.19). Find **H** inside and outside the rod.

*Solution.* Copper is weakly diamagnetic, so the dipoles will line up opposite to the field. This results in a bound current running antiparallel to $I$, within the wire, and parallel to $I$ along the surface (Fig. 6.20). Just how great these bound currents will be we are not yet in a position

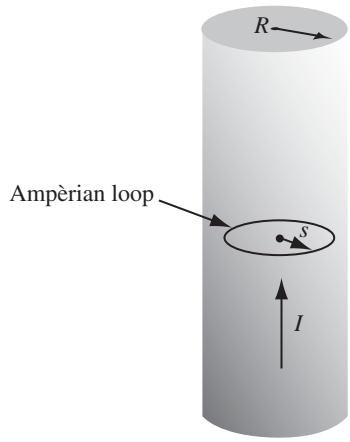



Fig. 6.19

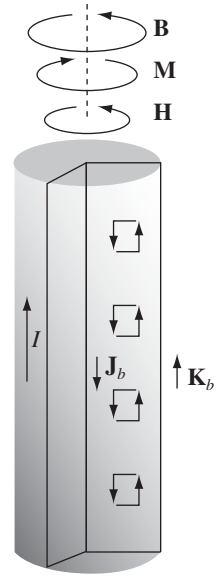



Fig. 6.20

to say – but in order to calculate **H**, it is sufficient to realize that all the currents are longitudinal, so **B**, **M**, and therefore also **H**, are circumferential. Applying Eq. 6.20 to an Ampèrian loop of radius $s < R$,

$$H(2\pi s) = I_{f_{\text{enc}}} = I\frac{\pi s^2}{\pi R^2},$$

so, inside the wire,

$$\mathbf{H} = \frac{I}{2\pi R^2}s\hat{\boldsymbol{\phi}} \quad (s \leq R). \tag{6.21}$$

Outside the wire

$$\mathbf{H} = \frac{I}{2\pi s}\hat{\boldsymbol{\phi}} \quad (s \geq R). \tag{6.22}$$

In the latter region (as always, in empty space) $\mathbf{M} = \mathbf{0}$, so

$$\mathbf{B} = \mu_0\mathbf{H} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\phi}} \quad (s \geq R),$$

the same as for a *non*magnetized wire (Ex. 5.7). *Inside* the wire, **B** cannot be determined at this stage, since we have no way of knowing **M** (though in practice the magnetization in copper is so slight that for most purposes we can ignore it altogether).

As it turns out, **H** is a more useful quantity than **D**. In the laboratory, you will frequently hear people talking about **H** (more often even than **B**), but you will never hear anyone speak of **D** (only **E**). The reason is this: To build an electromagnet you run a certain (free) current through a coil. The *current* is the thing you read on the dial, and this determines **H** (or at any rate, the

line integral of **H**); **B** depends on the specific materials you used and even, if iron is present, on the history of your magnet. On the other hand, if you want to set up an *electric* field, you do *not* plaster a known free charge on the plates of a parallel-plate capacitor; rather, you connect them to a battery of known *voltage*. It's the *potential difference* you read on your dial, and that determines **E** (or rather, the line integral of **E**); **D** depends on the details of the dielectric you're using. If it were easy to measure charge, and hard to measure potential, then you'd find experimentalists talking about **D** instead of **E**. So the relative familiarity of **H**, as contrasted with **D**, derives from purely practical considerations; theoretically, they're on an equal footing.

Many authors call **H**, not **B**, the "magnetic field." Then they have to invent a new word for **B**: the "flux density," or magnetic "induction" (an absurd choice, since that term already has at least two other meanings in electrodynamics). Anyway, **B** is indisputably the fundamental quantity (and it's the thing that goes in the Lorentz force law), so I shall continue to call it the "magnetic field." The quantity **H** has no sensible name: just call it "**H**."[10]

**Problem 6.12.** An infinitely long cylinder, of radius $R$, carries a "frozen-in" magnetization, parallel to the axis,

$$\mathbf{M} = ks\hat{\mathbf{z}},$$

where $k$ is a constant and $s$ is the distance from the axis; there is no free current anywhere. Find the magnetic field inside and outside the cylinder by two different methods:

(a) As in Section 6.2, locate all the bound currents, and calculate the field they produce.
(b) Use Ampère's law (in the form of Eq. 6.20) to find **H**, and then get **B** from Eq. 6.18. (Notice that the second method is much faster, and avoids any explicit reference to the bound currents.)

**Problem 6.13.** Suppose the field inside a large piece of magnetic material is $\mathbf{B}_0$, so that $\mathbf{H}_0 = (1/\mu_0)\mathbf{B}_0 - \mathbf{M}$, where **M** is a "frozen-in" magnetization.

(a) Now a small spherical cavity is hollowed out of the material (Fig. 6.21). Find the field at the center of the cavity, in terms of $\mathbf{B}_0$ and **M**. Also find **H** at the center of the cavity, in terms of $\mathbf{H}_0$ and **M**.
(b) Do the same for a long needle-shaped cavity running parallel to **M**.
(c) Do the same for a thin wafer-shaped cavity perpendicular to **M**.

[10] For those who disagree, I quote A. Sommerfeld's *Electrodynamics*, Academic Press, New York (1952), p. 45: "The unhappy term 'magnetic field' for **H** should be avoided as far as possible. It seems to us that this term has led into error none less than Maxwell himself." The **H** due to magnetization (as opposed to free currents) is sometimes called the **demagnetization field**, because it tends to oppose **M** (see, for example, Prob. 6.14).

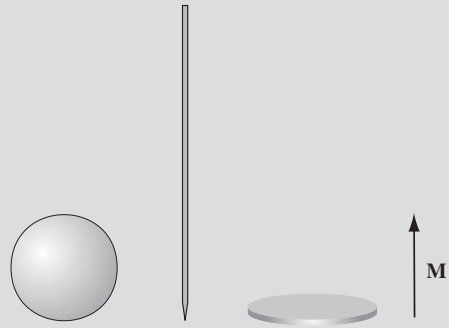



**Fig. 6.21**

Assume the cavities are small enough so $\mathbf{M}$, $\mathbf{B}_0$, and $\mathbf{H}_0$ are essentially constant. Compare Prob. 4.16. [*Hint:* Carving out a cavity is the same as superimposing an object of the same shape but opposite magnetization.]

### 6.3.2 A Deceptive Parallel

Equation 6.19 looks just like Ampère's original law (Eq. 5.56), except that the *total* current is replaced by the *free* current, and $\mathbf{B}$ is replaced by $\mu_0\mathbf{H}$. As in the case of $\mathbf{D}$, however, I must warn you against reading too much into this correspondence. It does *not* say that $\mu_0\mathbf{H}$ is "just like $\mathbf{B}$, only its source is $\mathbf{J}_f$ instead of $\mathbf{J}$." For the curl alone does not determine a vector field – you must *also* know the divergence. And whereas $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$, the divergence of $\mathbf{H}$ is *not*, in general, zero. In fact, from Eq. 6.18,

$$\boldsymbol{\nabla} \cdot \mathbf{H} = -\boldsymbol{\nabla} \cdot \mathbf{M}. \tag{6.23}$$

Only when the divergence of $\mathbf{M}$ vanishes (everywhere) is the parallel between $\mathbf{B}$ and $\mu_0\mathbf{H}$ faithful. In particular, there is (in general) no "Biot–Savart" law giving $\mathbf{H}$ in terms of the free current, and there is no vector potential for $\mathbf{H}$.

If you think I'm being pedantic, consider the example of the bar magnet – a short cylinder of iron that carries a permanent uniform magnetization $\mathbf{M}$ parallel to its axis. (See Probs. 6.9 and 6.14.) In this case there is no free current anywhere, and a naïve application of Eq. 6.20 might lead you to suppose that $\mathbf{H} = \mathbf{0}$, and hence that $\mathbf{B} = \mu_0\mathbf{M}$ inside the magnet and $\mathbf{B} = \mathbf{0}$ outside, which is nonsense. It is quite true that the *curl* of $\mathbf{H}$ vanishes everywhere, but the divergence does not. (Can you see where $\boldsymbol{\nabla} \cdot \mathbf{M} \neq 0$?) *Advice:* When you are asked to find $\mathbf{B}$ or $\mathbf{H}$ in a problem involving magnetic materials, first look for symmetry. If the problem exhibits cylindrical, plane, solenoidal, or toroidal symmetry, then you can get $\mathbf{H}$ directly from Eq. 6.20 by the usual Ampère's law methods. (Evidently, in such cases $\boldsymbol{\nabla} \cdot \mathbf{M}$ is automatically zero, since the free current alone determines the answer.) If

the requisite symmetry is absent, you'll have to think of another approach, and in particular you must *not* assume that **H** is zero just because there is no free current in sight.

### 6.3.3 Boundary Conditions

The magnetostatic boundary conditions of Section 5.4.2 can be rewritten in terms of **H** and the *free* current. From Eq. 6.23 it follows that

$$H^{\perp}_{\text{above}} - H^{\perp}_{\text{below}} = -(M^{\perp}_{\text{above}} - M^{\perp}_{\text{below}}), \tag{6.24}$$

while Eq. 6.19 says

$$\mathbf{H}^{\parallel}_{\text{above}} - \mathbf{H}^{\parallel}_{\text{below}} = \mathbf{K}_f \times \hat{\mathbf{n}}. \tag{6.25}$$

In the presence of materials, these are sometimes more useful than the corresponding boundary conditions on **B** (Eqs. 5.74 and 5.76):

$$B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0 \tag{6.26}$$

and

$$\mathbf{B}^{\parallel}_{\text{above}} - \mathbf{B}^{\parallel}_{\text{below}} = \mu_0(\mathbf{K} \times \hat{\mathbf{n}}). \tag{6.27}$$

You might want to check them, for Ex. 6.2 or Prob. 6.14.

**Problem 6.14.** For the bar magnet of Prob. 6.9, make careful sketches of **M**, **B**, and **H**, assuming $L$ is about $2a$. Compare Prob. 4.17.

**Problem 6.15.** If $\mathbf{J}_f = \mathbf{0}$ everywhere, the curl of **H** vanishes (Eq. 6.19), and we can express **H** as the gradient of a scalar potential $W$:

$$\mathbf{H} = -\boldsymbol{\nabla} W.$$

According to Eq. 6.23, then,

$$\nabla^2 W = \boldsymbol{\nabla} \cdot \mathbf{M},$$

so $W$ obeys Poisson's equation, with $-\boldsymbol{\nabla} \cdot \mathbf{M}$ as the "source." This opens up all the machinery of Chapter 3. As an example, find the field inside a uniformly magnetized sphere (Ex. 6.1) by separation of variables. [*Hint:* $\boldsymbol{\nabla} \cdot \mathbf{M} = 0$ everywhere except at the surface ($r = R$), so $W$ satisfies Laplace's equation in the regions $r < R$ and $r > R$; use Eq. 3.65, and from Eq. 6.24 figure out the appropriate boundary condition on $W$.]

## 6.4 Linear and Nonlinear Media

### 6.4.1 Magnetic Susceptibility and Permeability

In paramagnetic and diamagnetic materials, the magnetization is sustained by the field; when **B** is removed, **M** disappears. In fact, for most substances

the magnetization is *proportional* to the field, provided the field is not too strong. For notational consistency with the electrical case (Eq. 4.30), I *should* express the proportionality thus:

$$\mathbf{M} = \frac{1}{\mu_0}\chi_m \mathbf{B} \quad \text{(incorrect!)} \tag{6.28}$$

But custom dictates that it be written in terms of **H**, instead of **B**:

$$\boxed{\mathbf{M} = \chi_m \mathbf{H}.} \tag{6.29}$$

The constant of proportionality $\chi_m$ is called the **magnetic susceptibility**; it is a dimensionless quantity that varies from one substance to another – positive for paramagnets and negative for diamagnets. Typical values are around $10^{-5}$ (see Table 6.1).

Materials that obey Eq. 6.29 are called **linear media**. In view of Eq. 6.18,

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H} \tag{6.30}$$

for linear media. Thus **B** is *also* proportional to **H**:[11]

$$\mathbf{B} = \mu \mathbf{H}, \tag{6.31}$$

where

$$\mu \equiv \mu_0(1 + \chi_m). \tag{6.32}$$

The quantity $\mu$ is called the **permeability** of the material.[12] In a vacuum, where there is no matter to magnetize, the susceptibility $\chi_m$ vanishes, and the permeability is $\mu_0$. That's why $\mu_0$ is called the **permeability of free space**.

**Table 6.1.** Magnetic susceptibilities (unless otherwise specified, values are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010) and other references.

| Material | Susceptibility | Material | Susceptibility |
|---|---|---|---|
| *Diamagnetic:* | | *Paramagnetic:* | |
| Bismuth | $-1.7 \times 10^{-4}$ | Oxygen ($O_2$) | $1.7 \times 10^{-6}$ |
| Gold | $-3.4 \times 10^{-5}$ | Sodium | $8.5 \times 10^{-6}$ |
| Silver | $-2.4 \times 10^{-5}$ | Aluminum | $2.2 \times 10^{-5}$ |
| Copper | $-9.7 \times 10^{-6}$ | Tungsten | $7.0 \times 10^{-5}$ |
| Water | $-9.0 \times 10^{-6}$ | Platinum | $2.7 \times 10^{-4}$ |
| Carbon dioxide | $-1.1 \times 10^{-8}$ | Oxygen (liquid) (−200°C) | $3.9 \times 10^{-3}$ |
| Hydrogen ($H_2$) | $-2.1 \times 10^{-9}$ | Gadolinium | $4.8 \times 10^{-1}$ |

[11] Physically, therefore, Eq. 6.28 would say exactly the same as Eq. 6.29, only the constant $\chi_m$ would have a different value. Equation 6.29 is a little more convenient, because experimentalists find it handier to work with **H** than **B**.

[12] If you factor out $\mu_0$, what's left is called the **relative permeability**: $\mu_r \equiv 1 + \chi_m = \mu/\mu_0$. By the way, formulas for **H** in terms of **B** (Eq. 6.31, in the case of linear media) are called **constitutive relations**, just like those for **D** in terms of **E**.

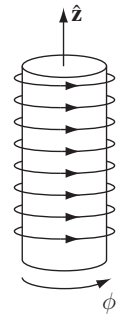



Fig. 6.22

**Example 6.3.** An infinite solenoid ($n$ turns per unit length, current $I$) is filled with linear material of susceptibility $\chi_m$. Find the magnetic field inside the solenoid.

*Solution.* Since **B** is due in part to bound currents (which we don't yet know), we cannot compute it directly. However, this is one of those symmetrical cases in which we can get **H** from the free current alone, using Ampère's law in the form of Eq. 6.20:

$$\mathbf{H} = nI\hat{\mathbf{z}}$$

(Fig. 6.22). According to Eq. 6.31, then,

$$\mathbf{B} = \mu_0(1 + \chi_m)nI\hat{\mathbf{z}}.$$

If the medium is paramagnetic, the field is somewhat enhanced; if it's diamagnetic, the field is slightly reduced. This reflects the fact that the bound surface current

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = \chi_m(\mathbf{H} \times \hat{\mathbf{n}}) = \chi_m nI\hat{\boldsymbol{\phi}}$$

is in the same direction as $I$, in the former case ($\chi_m > 0$), and opposite in the latter ($\chi_m < 0$).

You might suppose that linear media escape the defect in the parallel between **B** and **H**: since **M** and **H** are now proportional to **B**, does it not follow that their divergence, like **B**'s, must always vanish? Unfortunately, it does *not*;[13] at the *boundary* between two materials of different permeability, the divergence of **M** can actually be infinite. For instance, at the end of a cylinder of linear paramagnetic material, **M** is zero on one side but not on the other. For the "Gaussian pillbox" shown in Fig. 6.23, $\oint \mathbf{M} \cdot d\mathbf{a} \neq 0$, and hence, by the divergence theorem, $\boldsymbol{\nabla} \cdot \mathbf{M}$ cannot vanish everywhere within it.

Incidentally, the volume bound current density in a homogeneous linear material is proportional to the *free* current density:

[13] Formally, $\boldsymbol{\nabla} \cdot \mathbf{H} = \boldsymbol{\nabla} \cdot (\frac{1}{\mu}\mathbf{B}) = \frac{1}{\mu}\boldsymbol{\nabla} \cdot \mathbf{B} + \mathbf{B} \cdot \boldsymbol{\nabla}(\frac{1}{\mu}) = \mathbf{B} \cdot \boldsymbol{\nabla}(\frac{1}{\mu})$, so **H** is *not* divergenceless (in general) at points where $\mu$ is changing.

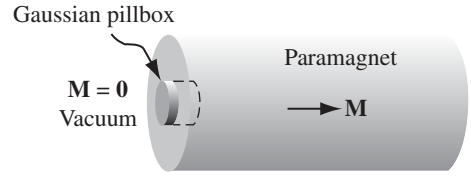



Fig. 6.23

$$\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M} = \boldsymbol{\nabla} \times (\chi_m \mathbf{H}) = \chi_m \mathbf{J}_f. \tag{6.33}$$

In particular, unless free current actually flows *through* the material, all bound current will be at the surface.

**Problem 6.16.** A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility $\chi_m$. A current $I$ flows down the inner conductor and returns along the outer one; in each case, the current distributes itself uniformly over the surface (Fig. 6.24). Find the magnetic field in the region between the tubes. As a check, calculate the magnetization and the bound currents, and confirm that (together, of course, with the free currents) they generate the correct field.

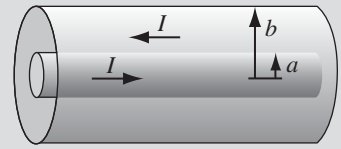



Fig. 6.24

**Problem 6.17.** A (free) current $I$ flows down a long straight wire of radius $a$. If the wire is made of linear material (copper, say, or aluminum) with susceptibility $\chi_m$, and the current is distributed uniformly, what is the magnetic field a distance $s$ from the axis? Find all the bound currents. What is the *net* bound current flowing down the wire?

! **Problem 6.18.** A sphere of linear magnetic material is placed in an otherwise uniform magnetic field $\mathbf{B}_0$. Find the field inside the sphere. [*Hint:* See Prob. 6.15 or Prob. 4.23.]

**Problem 6.19.** On the basis of the naïve model presented in Section 6.1.3, estimate the magnetic susceptibility of a diamagnetic metal such as copper. Compare your answer with the empirical value in Table 6.1, and comment on any discrepancy.

### 6.4.2 Ferromagnetism

In a linear medium, the alignment of atomic dipoles is maintained by a magnetic field imposed from the outside. Ferromagnets – which are

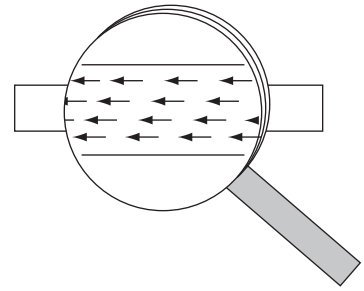


Fig. 6.25

emphatically *not* linear[14] – require no external fields to sustain the magnetization; the alignment is "frozen in." Like paramagnetism, ferromagnetism involves the magnetic dipoles associated with the spins of unpaired electrons. The new feature, which makes ferromagnetism so different from paramagnetism, is the interaction between nearby dipoles: in a ferromagnet, *each dipole "likes" to point in the same direction as its neighbors*. The *reason* for this preference is essentially quantum mechanical, and I shall not endeavor to explain it here; it is enough to know that the correlation is so strong as to align virtually 100 percent of the unpaired electron spins. If you could somehow magnify a piece of iron and "see" the individual dipoles as tiny arrows, it would look something like Fig. 6.25, with all the spins pointing the same way.

But if that is true, why isn't every wrench and skillet a powerful magnet? The answer is that the alignment occurs in relatively small patches, called **domains**. Each domain contains billions of dipoles, all lined up (these domains are actually *visible* under a microscope, using suitable etching techniques – see Fig. 6.26), but the domains *themselves* are randomly oriented. The household wrench contains an enormous number of domains, and their magnetic fields cancel, so the wrench as a whole is not magnetized. (Actually, the orientation of domains is not *completely* random; within a given crystal, there may be some preferential alignment along the crystal axes. But there will be just as many domains pointing one way as the other, so there is still no large-scale magnetization. Moreover, the crystals themselves are randomly oriented within any sizable chunk of metal.)

How, then, would you produce a **permanent magnet**, such as they sell in toy stores? If you put a piece of iron into a strong magnetic field, the torque $\mathbf{N} = \mathbf{m} \times \mathbf{B}$ tends to align the dipoles parallel to the field. Since they like to stay parallel to their neighbors, most of the dipoles will resist this torque. However, at the *boundary* between two domains, there are *competing* neighbors, and the torque will throw its weight on the side of the domain most nearly parallel to the field; this domain will win some converts, at the

[14] In this sense, it is misleading to speak of the susceptibility or permeability of a ferromagnet. The terms *are* used for such materials, but they refer to the proportionality factor between a *differential* increase in **H** and the resulting *differential* change in **M** (or **B**); moreover, they are not constants, but functions of **H**.